

# Sky Stream

Big Data Flight Delay Prediction System

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**From Reactive Reporting to Proactive Analytics**



# The Business Logic: Why This Matters

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- ⚠️ **The Problem:** Airlines operate reactively. Delayed flights cause domino effects of missed connections and blocked gates.
- 💡 **The Solution:** We built a proactive, real-time prediction engine to anticipate delays before they compound.
- 📈 **The Value:** Predicting arrival delays while flights are airborne allows airlines to re-route passengers, optimize crew schedules, and alert customers early.





# Enterprise Environment Setup

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## Containerization

Built a fully containerized microservices architecture using Docker Compose. We didn't just run local scripts; we built enterprise-grade infrastructure.



## Infrastructure as Code

Defined entire infrastructure through code. A single command spins up multiple isolated "virtual servers" that communicate over a custom Docker network.



## Reproducibility

Guarantees flawless setup. No manual installation of Java, Hadoop, or Spark required. Run the Docker file, and the ecosystem boots perfectly configured.



# The Big Data Infrastructure

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## Zookeeper & Kafka

The messaging nervous system. Handles high-throughput, fault-tolerant real-time data ingestion streams flawlessly.



## Hadoop HDFS

Our distributed Data Lake. NameNode and DataNode provide highly scalable, reliable storage for massive historical datasets.



## PySpark & Jupyter

The heavy-duty computing engine. Leverages distributed memory to process and train models on massive datasets rapidly.



# The Data Story

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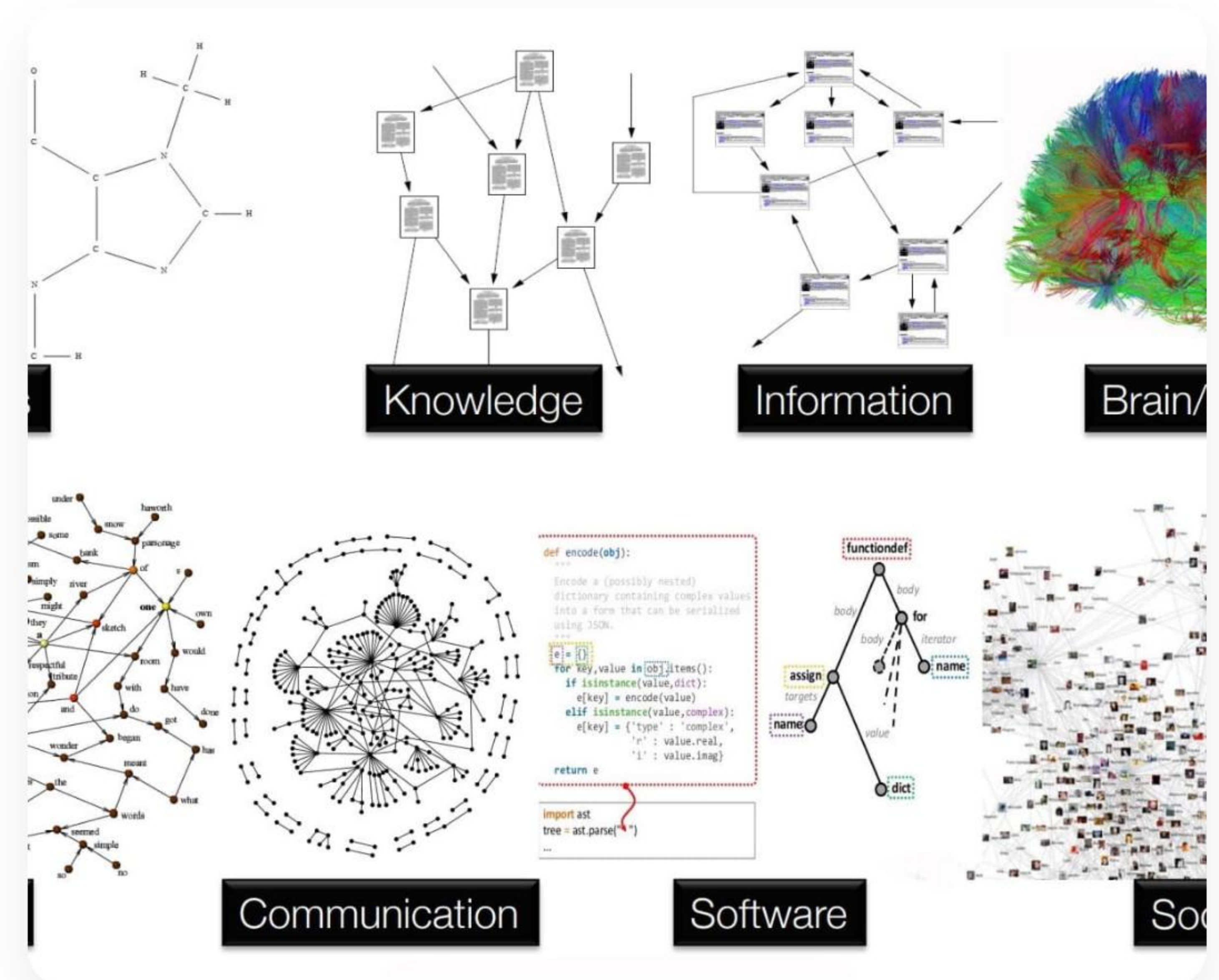
-  **The Source & Volume:** Utilized a massive, real-world 2024 Kaggle dataset comprising **over 7 Million** individual flight records.
-  **The Features:** Contains crucial metrics: scheduled departure/arrival times, carrier info, and historical delay lengths.
-  **Batch Pipeline:** Raw CSV data pushed into HDFS serving as historical ground-truth for training.
-  **Stream Simulation:** Custom script drips records line-by-line, simulating a live FAA or airline API feed.





# The Machine Learning Model

- 🧠 **The Algorithm:** XGBoost (Extreme Gradient Boosting) via PySpark MLlib. A powerful ensemble method for structured data.
- 🌲 **How It Works:** Builds hundreds of sequential decision trees. Each tree targets and corrects the errors of previous trees, learning complex non-linear relationships.
- ⚡ **Implementation:** Trained on 7M+ HDFS records, serialized, and saved as a model file for in-memory scoring of live flights.





# The Real-Time Engine (Lambda Architecture)

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## 1. The Producer (Trigger)

A Python script acts as the data source. It reads the raw data and publishes JSON messages directly into a Kafka topic named **raw-flight-data**.



## 2. Spark Streaming (Brain)

Acts as Consumer/Transformer. Listens to Kafka, cleans data, applies the XGBoost model, and pushes enriched data to a new **flight-predictions** topic.



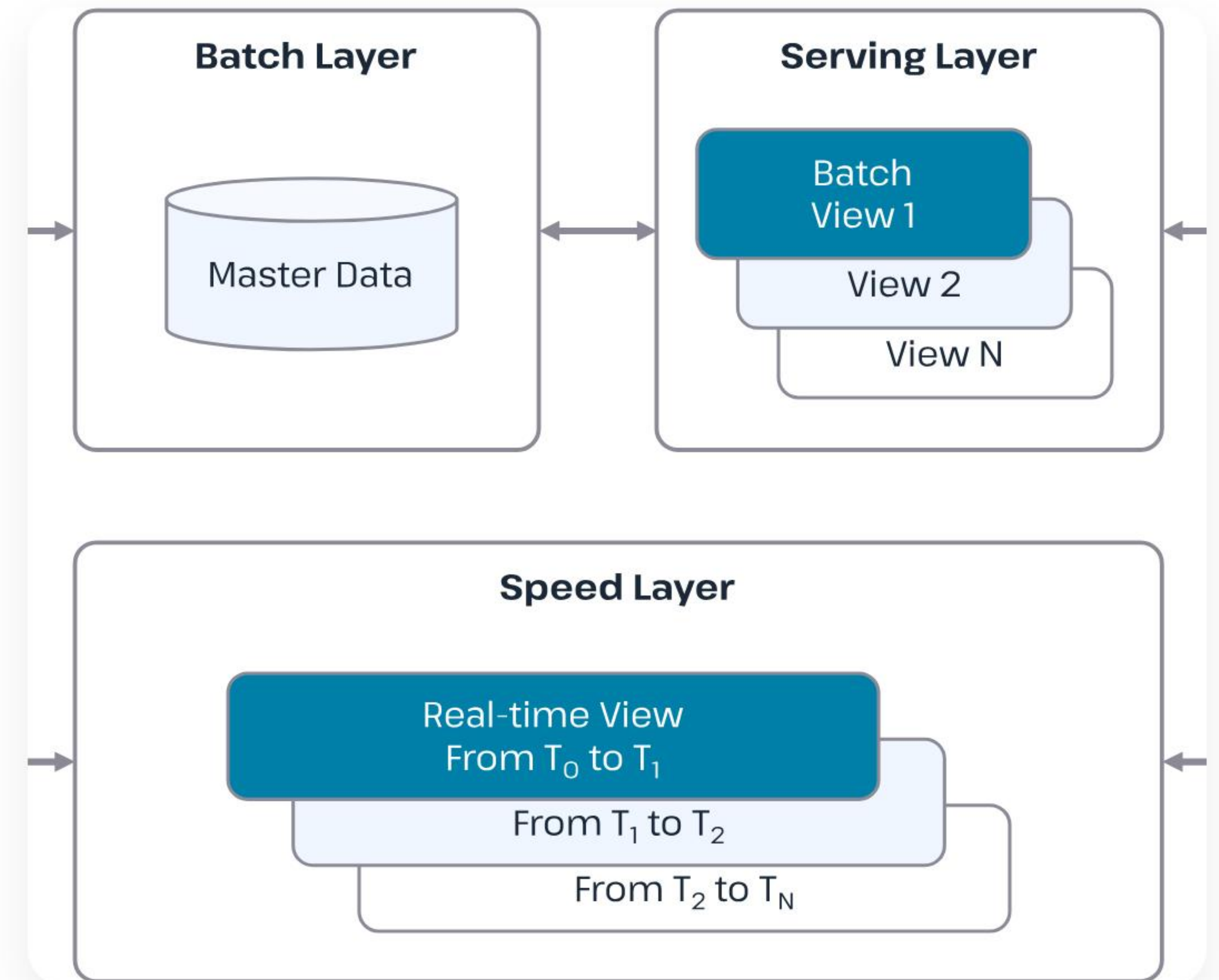
## 3. Streamlit (Face)

The final Kafka Consumer. Subscribes to predictions, pulling live data the second Spark scores it, updating UI metrics without page refreshes.



# Deep Dive: The Producer

-  **Data Ingestion:** Simulates a continuous stream of live flight telemetry.
-  **Python Integration:** Utilizes `confluent-kafka` library to establish a robust connection to the broker.
-  **JSON Formatting:** Formats complex tabular data into standardized JSON payloads for universal downstream compatibility.
-  **High Throughput:** Capable of handling thousands of messages per second, routing them to the raw-flight-data topic.





# Deep Dive: The Spark Brain

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## Continuous Processing

Spark Structured Streaming maintains an always-on connection to Kafka. As micro-batches arrive, schemas are instantly cast, and missing values are handled gracefully in-memory.



## Real-Time Transformation

Instead of writing to a slow database, Spark acts as a new Producer. It appends the XGBoost prediction delay and pushes the enriched payload to flight-predictions.



# Deep Dive: The Streamlit Face

- 👥 **End-User Visibility:** Bridges the gap between complex backend big data engineering and business operations.
- 🔄 **Live Polling:** The web app acts as the final consumer, subscribing directly to the Kafka output stream.
- 📊 **Dynamic Visualization:** Metrics, delay severity charts, and flight statuses update instantaneously as Spark finishes scoring.





# Any Questions?

Thank you for your time.

▶ Let's see Sky Stream in action!



# Image Sources

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[https://elements-resized.envatousercontent.com/elements-video-cover-images/files/237da759-c09e-4125-98d7-22c4504f3d30/inline\\_image\\_preview.jpg?w=500&cf\\_fit=cover&q=85&format=auto&s=258df99ba974845443087b16dc43195de51baf7d96011265ada90b7568c2c66d](https://elements-resized.envatousercontent.com/elements-video-cover-images/files/237da759-c09e-4125-98d7-22c4504f3d30/inline_image_preview.jpg?w=500&cf_fit=cover&q=85&format=auto&s=258df99ba974845443087b16dc43195de51baf7d96011265ada90b7568c2c66d)

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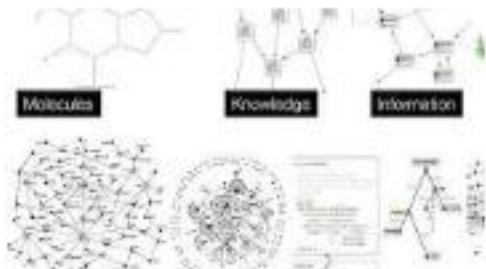
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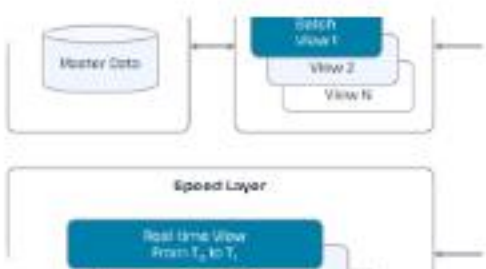
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